Question

Experiments show that the photo-induced reaction of 1,3-dimethyladamantane with Br_2 yields only one monobrominated product (stereoisomers not considered). Select the option containing all correct statements.

1. Theoretically, if stereoisomers are considered, there are a total of 8 monobrominated products

2. Theoretically, there are 5 monobrominated products that possess enantiomers

3. The experimentally obtained monobrominated product possesses enantiomers

A. 1,2,3

B. 1,2

C. 1,3

D. 2,3

E. 1

F. 2

G. 3

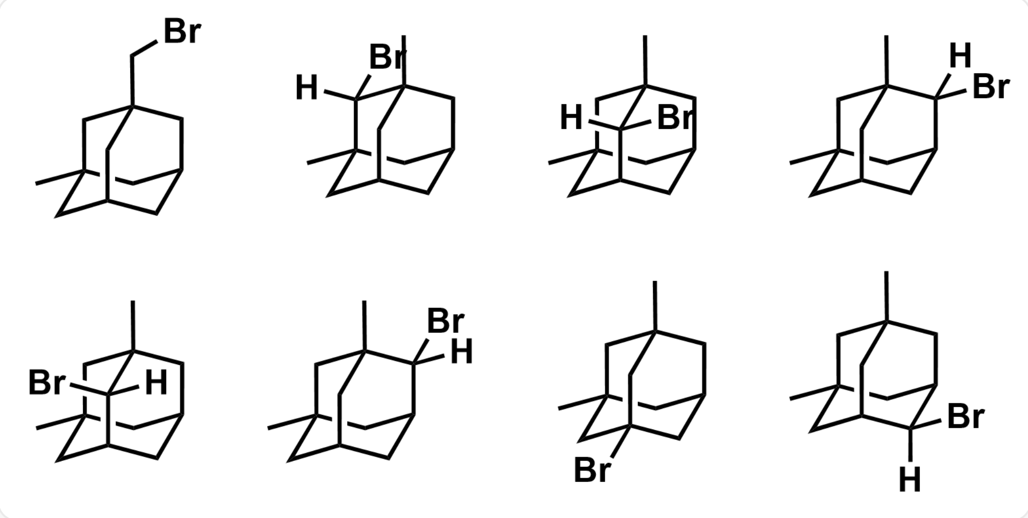
H. All of the above statements are incorrect.

Answer

Correct Answer: E

Detailed Explanation

First, analyze all possible monobrominated products of the alkane, considering stereoisomers. There are 8 kinds of monobrominated products as follows, statement 1 is correct.

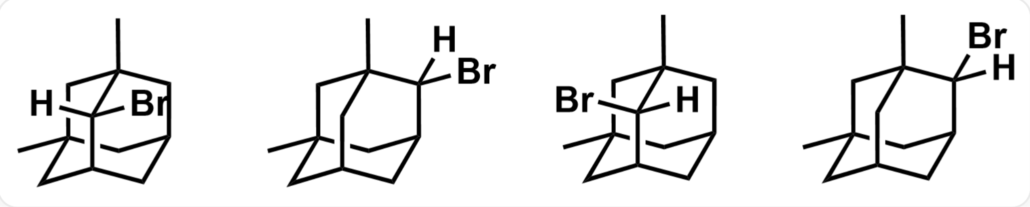


First row from left to right: C[C@@]1(C[C@@]2(CBr)C3)C[C@H]3C[C@H](C2)C1，C[C@@]12C[C@H]3C[C@](C1([H])Br)(C)C[C@@H](C2)C3，C[C@@]12[C@@](Br)([H])[C@H]3C[C@](C1)(C)C[C@@H](C2)C3，C[C@@]12C[C@H]3C[C@](C1)(C)C[C@@H](C2([H])Br)C3；Second row from left to right: C[C@@]12[C@](Br)([H])[C@H]3C[C@](C1)(C)C[C@@H](C2)C3，C[C@@]12C[C@H]3C[C@](C1)(C)C[C@@H](C2([H])Br)C3，C[C@@]12C[C@@]3(Br)C[C@](C1)(C)C[C@@H](C2)C3，C[C@@]12C[C@H]3C[C@](C1)(C)C[C@@H](C2)C3([H])Br

CHECKPOINT 1 PTS

There are 8 kinds of monobrominated products, statement 1 is correct

The following 4 monobrominated products do not have a mirror plane and have enantiomers, statement 2 is incorrect.

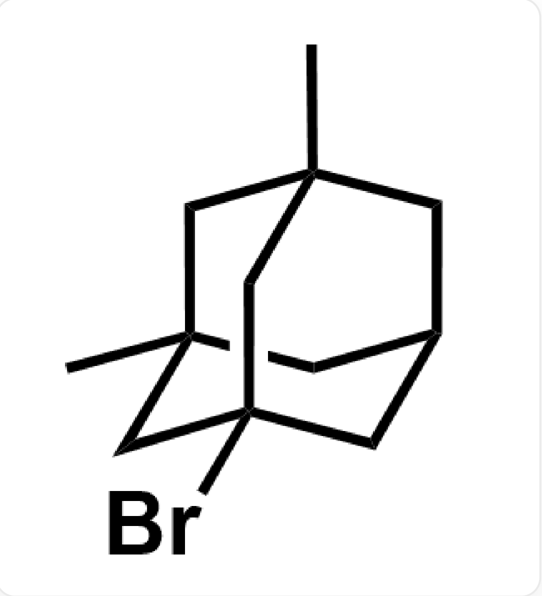


C[C@@]12[C@@](Br)([H])[C@H]3C[C@](C1)(C)C[C@@H](C2)C3，C[C@@]12C[C@H]3C[C@](C1)(C)C[C@@H](C2([H])Br)C3，C[C@@]12[C@](Br)([H])[C@H]3C[C@](C1)(C)C[C@@H](C2)C3，C[C@@]12C[C@H]3C[C@](C1)(C)C[C@@H](C2([H])Br)C3

CHECKPOINT 1 PTS

There are 4 monobrominated products with enantiomers, statement 2 is incorrect

The only monobrominated product obtained is generated from the most stable free radical. The free radical should be a tertiary free radical, and the product is as follows. It has a mirror plane and no enantiomers, statement 3 is incorrect.



C[C@@]12C[C@@]3(Br)C[C@](C1)(C)C[C@@H](C2)C3

CHECKPOINT 0.5 PTS

The experimentally obtained monobrominated product is generated from a tertiary free radical

CHECKPOINT 0.5 PTS

The experimentally obtained monobrominated product has no enantiomers

The answer is option E.